1. What is the primary objective of the Convolution Layer in a Convolutional Neural Network (CNN)?
A)To flatten the input image
B)To assign importance to various aspects/objects in the image
C) To reduce the spatial size of the input image
D)To perform element-wise multiplication of image arrays

2. Which of the following tasks is an example of computer vision?
A)Rescaling an image
B)Correcting brightness levels in an image
C)Object detection in images or videos
D)Changing tones of an image

3. How is resolution typically expressed?
A)By the number of pixels along the width and height, such as 1280x1024
B)By the brightness level of each pixel, ranging from 0 to 255
C)By the total number of pixels, such as 5 megapixels
D)By the arrangement of pixels in a 2-dimensional grid

4. What is the core task of image classification?
A) Identifying objects and their locations in images
B) Segmenting objects into individual pixels
C) **Assigning an input image one label from a fixed set of categories**
D) Detecting instances of real-world objects in images

5. What is the function of the Rectified Linear Unit (ReLU) layer in a CNN?
A) To reduce the image size for more efficient processing
B) To assign importance to various aspects/objects in the input image
C) **To get rid of negative numbers in the feature map and retain positive numbers**
D) To perform the convolution operation on the input image

6. Object detection and handwriting recognition are examples of tasks commonly associated with:
A) **Computer vision**
B) Image processing
C) Both computer vision and image processing
D) Neither computer vision nor image processing

7. What does the pixel value represent in an image?
A) Width of the pixel
B) **Brightness or color of the pixel**
C) Height of the pixel
D) Resolution of the pixel

8. In the byte image format, what is the range of possible pixel values?
A) 0 to 10
B) 0 to 100

C) 0 to 1000
D) **0 to 255**

9. In a grayscale image, what does the darkest shade represent?
A) Total presence of color
B) **Zero value of pixel**
C) Lightest shade of Gray
D) Maximum pixel value

10. In an RGB image, what does a pixel with an intensity value of 0 represent?
A) Full presence of color
B) **No presence of color**
C) Maximum brightness level
D) Minimum brightness level

11. Assertion: Object detection is a more complex task than image classification because it Involves identifying both the presence and location of objects in an image.
Reasoning: Object detection algorithms need to not only classify the objects present in an Image but also accurately localize them by determining their spatial extent.
Select the appropriate option for the statements given above:
A) Both A and R are true and R is the correct explanation of A
B) Both A and R are true and R is not the correct explanation of A
C) A is true but R is false
D) A is False but R is true

12. Assertion: Grayscale images consist of shades of Gray ranging from black to white, where each pixel is represented by a single byte, and the size of the image is determined by its height multiplied by its width.
Reasoning: Grayscale images are represented using a three intensities per pixel, typically ranging from 0 to 255.
Select the appropriate option for the statements given above:
A) Both A and R are true and R is the correct explanation of A
B) Both A and R are true and R is not the correct explanation of A
C) A is true but R is false
D) A is False but R is true